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APPLIED RESEARCH

Machine Learning for Real-Time Fuel Consumption Prediction and Driving Profile Classification Based on ECU Data

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ABSTRACT Data extracted directly from a vehicle's electronic control unit (ECU) play a crucial role in the automotive industry because they contain valuable information from the engine and electronic parts. These data have the potential to enable compliance analysis, detect faults and errors, and guarantee driver and car safety as well as product quality. Among the possible uses of the data from the ECUs, driving profile analysis and fuel consumption prediction stand out, which enable analyses for insurers and transportation companies, and help to reduce fuel consumption and greenhouse gases, in addition to providing feedback to the driver. In this work, we apply machine learning algorithms to real data from an engine ECU to examine the driver's driving behavior and accurately classify their fuel efficiency. Moreover, we develop regression models that predict fuel consumption for vehicles in operation. To ensure the effectiveness of our models, we carefully select variables strongly correlated with fuel consumption using a feature selection process. Compared to related works, both our profile classification results in precision, recall, and accuracy metrics, and our regression models result in the metrics of mean square errors, mean absolute error, and coefficient of determination, which are superior or similar. Notably, our algorithms exhibit lower computational costs and enable real-time analysis by utilizing a cloud server.

INDEX TERMS Machine learning, driving profile analysis, fuel consumption prediction, electronic control unit (ECU).

I. INTRODUCTION

To guarantee the efficiency and performance of a vehicle during operation, it is crucial to monitor the combustion engine to avoid problems that can cause damage, decrease the lifetime, and increase its gas emissions and fuel consumption. By reading data from the vehicle's Electronic Control Unit (ECU), it is possible to monitor different engine variables at runtime during vehicle development phases, thus ensuring and improving its quality [4].

In the current automotive sector, most of the market value of an automobile is related to the technology used by the industry, and 60% of its executives believe that companies are well prepared for Industry 4.0, such as artificial intelligence [19]. Industry 4.0 represents a modern automation and data communication paradigm in manufacturing and related industries, connecting technologies like cyber-physical systems, big data analytics, IoT, and cloud computing to develop smart applications [2]. In this sense, new data-based strategies are being applied in the study of motors and their parts, to improve interactivity with the driver, provide feedback, bring relevant information from the car, and boost industrial processes, such as calibration, gas emission control, and consumption reduction [7].

Machine learning (ML) algorithms gained notoriety in numerous research and studies due to their ability to detect

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anomalies, faults, or changes in the behavior pattern of the engine or a set of variables [35], [37], [41]. Among the phenomena that can be analyzed from the ECU data using ML, two are the classification of driving behavior [33] and fuel consumption prediction [17]. Driver profile monitoring can identify vehicle thefts, which is interesting for insurers and transport companies, in terms of security and personalized insurance plans [13], [14]. Selling data to insurance companies is a growing market, 43% of automakers will sell driver and vehicle data to insurance companies in a short time [19]. Furthermore, profile analysis can define patterns to reveal improper driving conduct that may cause traffic accidents or signs of serious engine failure, making preventive actions possible to avoid more significant dangers [11]. In addition, it provides the opportunity to monitor the main parameters and variables that affect the control of gas emissions, substance leakage, and excessive vehicle consumption [15], [30], [38].

Fuel consumption prediction has potential benefits, whether for economic or environmental purposes, mostly in works related to transport [17], and has been using ML to develop predictive models in recent years [11], [16], [42]. The benefits of AI-based models include the minimization of fuel consumption due to the drop in gas emissions of toxic chemical compounds to the environment. Moreover, fuel consumption prediction can help in the calibration of the engine and its systems, a complex process that requires many tests and physical prototypes of the vehicles [7]. This is an expensive and time-consuming step in the process of building/testing new engines, which makes it difficult to comply with legislation. ML has several strengths over traditional methods, including its ability to handle large amounts of data and complex relationships between variables [45]. ML algorithms can also provide a more accurate prediction of real-world fuel consumption based on a broader range of variables than traditional linear regression models [26] and can reduce the need for physical testing, without the need for new calibrations [25].

Related works have been proposed to identify and/or classify driving conduct by labeling driving habits as safe or unsafe [21], applying cluster algorithms to classify the driver as eco-friendly or not [28], [37], and verifying the impact of monitoring on fuel consumption [23], [35]. For fuel consumption regression, studies proposed models based on driving behavior data collected by smartphones and OBD terminals [42], applied Support Vector Machine [11], and developed Deep Neural Network models [16] to predict fuel consumption. In general, the works obtained good results, however, with issues that can be improved. Whether in the sets of variables used, which may not be present in all vehicles [21], not associating the models with factors of intrinsic fuel consumption, such as acceleration [37], in the training data, which the authors did not collect [21], [41], using equipment with lower sampling frequencies [42], such as OBD terminals, developing algorithms that require more computational power [16], or in the execution time, that not occur at runtime [23].

Our work deals with two approaches related to fuel consumption. The first one aims to classify drivers' driving profiles, whether economical or not, using three ML algorithms (one unsupervised and two supervised). The second aims to predict fuel consumption, applying three ML models of regressors. The main difference in our work when compared to related work is twofold. First, we acquire real-time data directly from the car's ECU and send it to a cloud server. Second, we apply ML models on the server as data arrives at runtime, providing instant feedback to engineers or users. In summary, we make the following contributions:

1) We implemented and compared three ML-based algorithms (K-Means, Logistic Regression, and XGBoost), representing different classes of classifiers, to categorize driving behavior as fuel-friendly or not. Likewise, [28] compared algorithms and obtained similar results; however, implemented computationally more expensive algorithms, such as neural networks [8];
2) We developed and compared three algorithms (Ridge Regression, Support Vector Regression, and XGBoost) for fuel consumption prediction. [16], [42] also worked on predicting consumption; however, they used neural networks and smartphones, requiring more resources and greater computational power [8];
3) We trained all models with real data captured from a test vehicle and compared the performance of the algorithms with changes in driving and environment. Our results indicate that we can classify the driving behavior with up to 100% accuracy, recall, and precision. Besides predicting consumption with an average of the square of errors between predicted and measured values of 0.28 (l/h^2), mean absolute error of 0.23 (l/h), and coefficient of determination of 0.99;
4) We conducted a comprehensive literature review and compared existing works regarding resources used, techniques applied, and results obtained, summarizing this in a table.

This study extends an article presented at ISIE 2023, which initially addressed the driver's driving profile classification [5]. It has been expanded to include the regression analysis of instantaneous fuel consumption and its outcomes, aiming to provide a more comprehensive understanding of fuel consumption and its correlation with the vehicle's behavior.

The paper is organized as follows. Section II reviews the related works. Section III summarizes the data acquisition methodology we used in this work. Section IV describes the used algorithms and their parameters. Section V presents the achieved results. Finally, Section VI concludes the paper.

II. RELATED WORK

ML algorithms have been explored in automotive-related research recently, given their ability to analyze data for

pattern recognition, whether analyzing driver [36] or vehicle behavior [10], mainly because of their relevance in developing fuel consumption models based on real-world driving data, which can improve the accuracy of engine-related parameters and reduce the need for physical tests [7], [25], [26]. Moreover, task offloading emerges as a promising strategy for addressing the computational demands of vehicular applications, given the constraints imposed by limited onboard computing capabilities [39]. Consequently, ML algorithms developed in the cloud have an even greater potential to assist with vehicle applications, as they have exponentially higher computational power compared to onboard vehicle computers and they can provide real-time analysis processing during vehicular operations

In the following subsections, we review the literature considering two main related topics: (i) classification of the driver's driving profile in terms of fuel consumption [28], [41], [44]; and (ii) regression models to estimate the fuel consumption, including analyses of related factors (environmental, traffic, driving behavior) [22], [24], [32]. Compared to human-designed analyses of engine data, ML algorithms speed up the processing with a greater amount of data [29]. The ML Classifiers section presented below was extracted from [5], which is the article being expanded in this study.

A. ML CLASSIFIERS

Driver behavior is one of the biggest factors that determine vehicle fuel consumption and, therefore, driver profile analysis can make a difference in the car's economy when it is coupled with accurate feedback. The difference in fuel consumption reaches up to 30%, depending on the driver [23]. The works discussed below explored different techniques in the analysis of the driver's profile and its classification.

The classification of driver styles and behaviors can be made from vehicle-obtained data and, in some occasions, it is desirable to make the classifier decision based on a short time window. [44] implemented an algorithm to classify new observed drivers in different styles of driving, inferring the style and comparing it to existent classes determined by previous analysis and clustering. However, the obtained accuracy results varied between 70-90%. Using the relationship between acceleration and vehicle speed, [21] developed a classifier to label driving habits as safe or unsafe. The classification tools were developed using the Support Vector Machine and feedforward neural network techniques, and the results obtained show an accuracy of around 90% in both techniques. The models were trained using data from an open dataset, while our work collected all the data used.

Vehicle consumption and pollutant emissions are related to the driving profile. With this premise, [28] created an interface to gather car sensor data and use it in algorithms, classifying the driver behavior as pro or non-eco-friendly, comparing the results and the performances. A K-means algorithm was selected for clustering and classifying data and another three machine learning algorithms were implemented and compared (logistic regression, Support Vector Machine, and Random Forest) as well as two deep learning methods (Multilayer Perceptron and the Recurrent Neural Network algorithms). All explored algorithms have shown similar results, getting near 100% accuracy, being the more explicit difference noted in the execution time, whereas the Deep Learning algorithms show far slowest times. The results are similar to ours, however, we did the classification directly with K-means, and with two other algorithms with lower computational costs.

Reference [35] used in-vehicle data recorders-based feedback to explore its influence on driving behavior in three phases. First, without any feedback, second with feedback limited to the worst drivers, and last with feedback provided to all drivers. A fuel consumption reduction between 2-6.5% was observed from the first to the second phase, and between 3-10% from the first to the third phase, proving that the existence of monitoring and feedback helps and encourages drivers to improve their attitudes when driving, influencing fuel consumption and safety. In the same way, [23] installed driver performance measurement equipment on 12 buses to test and confirm three hypotheses: (i) the driver-independent factor has a considerable effect on the fuel consumption of buses; (ii) the effect of these factors can be isolated to allow a fair comparison of drivers based on fuel consumption; and (iii) fuel consumption can be used as a performance indicator in an eco-driving incentive system for drivers if driver-independent factors are isolated. The study concluded that just the awareness that fuel consumption is monitored makes drivers drive more economically and that differences in driver savings percentages are based on driver behavior and not on external factors. This study does not perform real-time analysis, unlike ours, which monitors the driver's profile and analyzes data while the vehicle is running.

Other studies developed adaptive driving-style-oriented equivalent consumption minimization strategies for hybrid electric vehicles [40] and measured the effect of driving style on fuel consumption and traffic flow emissions based on field data [41]. Reference [29] observed the impact of driver behavior on fuel consumption by classifying, evaluating, and predicting consumption using ML methods, such as neural networks, and visual information beyond the vehicle-gathered data, getting an accuracy of 80% to 83%. Nonetheless, we obtained superior metrics working only with vehicle features, reaching 100% accuracy through algorithms that require less computational power (Logistic Regression and Gradient Boosting).

Table 1 summarizes the strategies used and the variables monitored in related works. In the present work, clustering and classification algorithms were developed and compared, and the results point to a correct division between the measured driving data, separating economical and non-economical drivers with accuracy, precision, and recall around 100%, varying between the experiments done. In general, we developed a work that combined the best practices noted above, correcting flaws such as data

TABLE 1. Summary of related works that use classification techniques. Source: [5].

Source	Features	Techniques	Metrics
[21]	Vehicle speed, engine speed, engine load, throttle position, steering wheel angle, and brake pedal pressure	Support vector machine and feed-forward neural network	Accuracy(%): 90
[28]	Altitude, bearing, speed, RPM, intake air temperature, engine temperature, throttle position, fuel, and engine runtime	Logistic regression, support vector machine, random forest, multilayer perceptron, and long short-term memory (recurrent neural network)	Accuracy(%): 98.2, 100, 100, 99.8, and 100 (respectively)
[23]	Fuel consumption, braking, RPM, speed, distance, and time operating	Correlation analysis and hypothesis testing	Not applicable
[35]	Braking, lateral acceleration, speeding, safety-related events, and fuel consumption	Feedback based on In-Vehicle Data Recorders	Reduction of 8% in safety incidents, and 3–10% in fuel consumption with feedback, a larger reduction obtained for large vehicles
[41]	Position, speed, acceleration, time-to-collision, time headway and safety margin	K-means (classifying), LSTM (prediction/simulation), and correlation analysis (fuel consumption/traffic settings)	Pearson Correlation Coefficient (Positive/Negative lead speed change): fuel consumption rate: 0.997/-0.997, fuel emission rate: 0.999/-0.965, and fuel emission of aggressive drivers: -0.99/0.9932
[44]	Vehicle positions, velocity, and acceleration	Short-term observations algorithm	Based on K-folds cross-validation metrics, accuracies between 70-90% were found
[40]	Speed, throttle position, and acceleration	Density Estimation and Entropy Theory (probability estimation) and Rate Comparison	Fuel consumption Reduction (%): aggressive driving style: 1.8 - 4 and moderate driving style: 2.5 - 12
[29]	Speed, acceleration, brake pedal position, throttle position, and steering	Spectral Clustering Algorithm (clustering) and RNN - LSTM (predicting)	Clustering accuracy rate (%): 79.31 and Prediction Accuracy (%): 83.6
[5]	Speed, acceleration, engine speed, engine temperature, engine air load, torque, throttle valve position, accelerator pedal position, battery voltage, clutch pedal position	K-means (clustering), Logistic Regression, and XGBoost (classifiers)	Accuracy (%), precision (%), and recall (%): 93.55, 100, and 91.70 in clustering; 100, 100, and 100 in regression; and 100, 100, and 100 in boosting.

inconsistency, less accurate results, communication with the cloud, and real-time execution.

B. ML REGRESSORS

To predict a specific value in the form of classification or regression, a Support Vector Machine (SVM) algorithm can be used. Reference [11] developed two methods applying SVM, one based on Revolution Per Minute-Throttle Position Sensor (RPM_TPS) and another based on Vehicle Speed-Mass Air Flow (VS_MAF), both based on a legacy dataset containing Onboard Diagnostic (OBD) data with 18 variables. The proposed model consists of four phases: data pre-processing, feature weighting, feature selection, and SVM prediction model. The performance of the proposed model is evaluated using the R^2 metric, where the VS_MAF had an R^2 of 0.97, and RPM_TPS had 0.96. More metrics could have been applied to evaluate the proposed model, as we did in our work.

Reference [24] proposed a model using support vector regression (SVR) to predict fuel consumption by adopting a steady-state estimation and transient correction, using a dataset originated from the D3 database of the Advanced Powertrain Research Facility at Argonne National Laboratory [3]. They constructed a model divided into two modules, first to reflect the transient state of the vehicle, using Vehicle Speed and Vehicle Acceleration, and second to reflect the impact of vehicle inclination and load on fuel consumption, using Engine RPM and Engine Torque. SVR was used to predict fuel consumption and, to accurately classify the driving conditions, the modeling data was analyzed using the Gaussian Mixture Model (GMM) [31]. The results show the model's Root Mean Squared Error (RMSE) of 0.2137, the Mean Absolute Percentage Error (MAPE) of 13.6896%, and the Mean Absolute Error (MAE) of 0.1526. One of the strengths of our work compared to this one is the use of data acquired and validated by us, with the application of models in real-time while the vehicle is in operation.

Predicting vehicular fuel consumption by driving cycles can be helpful in economy strategies, especially for optimizing the Energy Management Systems of Hybrid Electric Vehicles (HEV). Based on that premise, [32] generated twelve personalized driving cycles, using driver demographic information, and driving style category to predict fuel consumption. The prediction was made through physical and polynomial models, for both traditional and hybrid cars, using an open dataset containing real-world GPS-based driving records. Results show that in conventional combustion vehicles, only the calm driver cycles showed consumption lower than the standards. In comparison, all personalized HEV cycles had lower consumption rates on the urban ones, besides an overall better performance than the conventional vehicles. Prediction analysis on the optimized

HEV shows that adjustments based on driving cycles can save up to 12% of fuel usage, with higher savings for urban scenarios. The authors based part of their study on simulations, creating driving cycles, while in our approach, we capture real vehicle data on predefined routes and driving profiles.

Reference [17] compare different ML regression models in consumption prediction using experimental data collected in heavy-duty trucks with distinct driving styles, driving through various routes and conditions. They applied the Recursive Feature Elimination feature selection method and selected engine load, vehicle speed, and engine speed to train the model. A feedforward Artificial Neural Network (ANN), Multiple Linear Regression (MLR), and Random Forest (RF) algorithms were implemented and tested for comparison. ANN was capable of an R^2 score of 0.78, showing slightly better results than the other models related to cumulative fuel consumption, with RMSE of 0.0025 L, while MLR showed 0.73 R^2 and 0.0029 RMSE, and RF had a 0.72 R^2 and RMSE of 0.0030.

With another approach, [42] proposed three fuel consumption prediction models based on driving behavior data collected by smartphones and OBD terminals. The developed models were RF, SVM, and Back Propagation (BP) Neural Network. The data used in the article was collected from taxis and mobile phone terminals with a sampling interval of 1 second, 75% of the collected data was randomly selected as training samples, and the remaining data were used as test samples to evaluate the accuracy of the developed models. To evaluate the accuracy and efficiency of the models they used RMSE, mean absolute percentage error (K), coefficient of determination (R^2), and model running time. The RF model obtained an RMSE of 0.783, K of 0.069, R^2 of 0.635, and a running time of 0.140 seconds. The SVM model obtained an RMSE of 0.888, K of .073, R^2 of 0.519, and a running time of 0.933 seconds. The BP Neural Network model obtained an RMSE of 0.872, K of 0.075, R^2 of 0.547, and a running time of 0.724 seconds. Additionally, the article highlights that using smartphones to collect data can reduce costs associated with installing OBD devices, something that does not have much impact because the use of OBD in vehicles is mandatory. Compared to our work, [17] made a limited selection of features, while we applied several different algorithms to find the common variables most closely related to fuel consumption. Reference [42], however, used ODB data, which has a reduced sampling frequency when compared to the acquisition directly from the ECU, in addition to using data from the cell phone, which creates a dependence on one more extra component to those found in the vehicle.

Reference [45] focused on the performance of MLP models in predicting fuel consumption using 12 variables (7 quantitative and 5 qualitative) as input parameters. The methodology involves collecting data on the technical parameters of passenger cars fabricated between 2010 and 2020, creating a database for training the ANN and setting input and output variables based on a literature review. The data used was obtained from international tests conducted by automotive factories. The work reports a high rate of training, testing, and validation for the MLP model with an accuracy of 0.93-0.95, RMSE values of 0.40–1.15, MSE of 0.2249-0.4010, and MAPE of 5%–11.5%. While the work reports high accuracy for the MLP model in predicting fuel consumption, it does not provide a direct comparison between MLP and other ML models. Another work applied an MLP model for predicting real-world fuel consumption rates of light-duty vehicles [22]. The model considers input parameters like external environmental factors, vehicle company manipulations, and drivers' habits. To train and test the MLP model, the authors used a dataset of real-world fuel consumption rates for light-duty vehicles, splitting it into training and testing sets, by 70% and 30% respectively, they also performed k-fold cross-validation. Overall, they tried to demonstrate that the article provides a valuable contribution to the field of ML to predict real-world fuel consumption rates, but it does not show any comparison or numerical evaluation that guarantees this in comparison to other works found in the literature. Contrary to our study, where this was a point on which we worked a lot, in addition to the intense feature selection process that we carried out, something that was also superficial in these works.

Reference [16] develop an estimation model of instantaneous fuel consumption in vehicles using data measured by smartphone sensors. The proposed model consists of a Deep Neural Network, using an approach called soft sensors, that returns estimations of measure through modeling based on other sensor's measures. The fuel consumption data were obtained from the OBD via the vehicle's CAN-bus. To construct the estimation model, the authors compared two existent types of Deep Neural Networks: Long Short-Term Memory (LSTM) and Nonlinear Autoregressive with Exogenous Inputs (NARX) networks. The second showed smaller errors and was chosen for the final model test where they performed cross-validation, and compared several training algorithms, to test the optimization methods that better perform in fitting the data, including different gradient-based, population-based, and large-scale population-based methods. In general, all of them showed good results, with average errors below 1 km/l, but the contrast-based Fruit Fly Optimization presented better scores, with median squared errors down to 0.35 km/l, with variance between predicted and measured data showed scores up to 96%. Finally, they tested the model with different sets of available features to find the set that best predicts consumption. Although, the complete set showed far better scores, which may indicate a very strong relationship between the model and the dataset, not necessarily a relationship between the variables and fuel consumption. In our work, we were able to validate the relationship between the variables and fuel consumption, with a thorough feature selection stage and applying the models to several different data sets, including run-time driving data.

TABLE 2. Summary of related works that use regression techniques.

Source	Features	Techniques	Metrics
[11]	Vehicle's work period, latitude, longitude, altitude, barometric pressure, engine coolant temperature, fuel level, ambient, air temperature, engine rotation per minute, intake manifold pressure, amount of air entering the engine, air intake temperature, speed, ECU signaling response according to the chances of oxygen levels, throttle position, timing advance, air/fuel ratio	Support vector machine	The vehicle Speed-Mass Air Flow equation and Revolution Per Minute-Throttle Position Sensor equation had an R^2 of 0.97 and 0.96, respectively.
[24]	Vehicle speed, vehicle acceleration, engine speed, and engine torque	Support vector regression	MAPE less than 14%, MAE less than 0.16 cc/s, R^2 0.97, and RMSE 0.1164 cc/s
[32]	Vehicle mass, aerodynamic drag coefficient, vehicle frontal area, rolling resistance coefficient, tire radius, engine-rated power, torque, continuous power output, max torque output, max speed, battery-rated voltage, and battery capacity	Used physical models for estimating fuel consumption	Adjusts based on driving cycles can save up to 12% of fuel usage, with higher savings for urban scenarios
[17]	Engine load, vehicle speed, and engine speed	Feedforward artificial neural network, multiple linear regression, and a random forest	Artificial Neural Network, Multiple Linear Regression, and Random Forest: R^2 0.78 and RMSE of 0.0025 L, R-square 0.73 and RMSE 0.0029, and R-square 0.72 and RMSE 0.0030 L, respectively.
[42]	Vehicle speed, engine load, coolant temperature, fuel consumption rate, and GPS location	Random Forest, Support Vector Machine, and Back Propagation Neural Network	Random Forest, Support Vector Machine, and Back Propagation Neural Network: RMSE of 0.783, K of 0.069, R^2 of 0.635, and a running time of 0.140 seconds; RMSE of 0.888, K of 0.073, R^2 of 0.519, and a running time of 0.933 seconds; RMSE of 0.872, K of 0.075, R^2 of 0.547, and a running time of 0.724 seconds; respectively
[45]	Cubic capacity, the number of cylinders, the number of valves, maximum power, maximum torque, compression rate, kerb weight of vehicle, type of engine, fuel injection, type of charge, gearbox, and drivetrain	Multilayer Perceptron	Accuracy of 0.93-0.95, RMSE of 0.40–1.15, MSE of 0.2249-0.4010, and MAPE of 5%–11.5%.
[22]	The specific inputs used in the model are not explicitly listed but include external environmental factors, vehicle company manipulations, and drivers' driving habits	Multilayer perceptron	Not applicable, evaluation metrics were not used.
[16]	Time, GPS position (latitude, longitude, altitude), speed, acceleration (longitudinal, lateral, vertical), and the number of visible satellites	NARX Recurrent Neural Network	MSE of 0.43 kpl, correlation (predicted/measured) of 0.96
[5]	Speed, acceleration, engine speed, engine temperature, engine air load, torque, throttle valve position, accelerator pedal position, battery voltage, clutch pedal position	XGBoost, Ridge Regression, and Support Vector Regression	MSE $(l/h)^2$, R^2, and MAE (l/h): 0.37, 0.99, and 0.25 for XGBoost; 0.28, 0.99 and 0.23 for Ridge; and 0.38, 0.99, and 0.30 for Support Vector Regression, respectively.

Table 2 summarizes the methods and variables used and the results obtained in related works. In our work, regression algorithms were developed and compared, and the results point to a correct prediction of fuel consumption, with metrics MSE $(l/h)^2$, R^2, and MAE (l/h) of 0.28, 0.99, and 0.23, respectively, varying between the models and experiments carried out. Overall, our work merged the best practices pointed out above, improving the feature selection, using data exclusively from the vehicle's ECU, and avoiding data inconsistency, so that the results have been improved or maintained, but communicating with the cloud, running in real-time and without the need to add more devices.

III. DATA ACQUISITION METHODOLOGY

Industry 4.0 technologies enable real-time monitoring of drivers, vehicles, and engine calibration parameters, as well

as producing a significant amount of data for analysis [2]. This visibility allows automotive companies to simplify their test pipeline, reduce costs, and improve overall operational efficiency.

Therefore, this study employed the acquisition system described by [5] and [9]. To ensure the integrity of the analyses proposed in this research, authentic data was collected from a Renault Sandero 2019, provided by Renault do Brasil, using custom acquisition hardware connected to the engine's ECU as described by [4]. Importantly, we exclusively rely on real vehicle data, avoiding simulated or public datasets, a distinctive feature from similar studies.

With a graphical user interface, the acquisition system allows the configuration and targeting of ECU variables. The chosen variables are standard across vehicle models, making our approach broadly applicable. Using 4G communication with a cloud server, we demonstrate real-time data processing and application to machine learning models or other treatments, as shown in Figure 1. ML algorithms (workflows), implemented in Python 3.8, are triggered based on marked variables. Acquired data is stored in a database for subsequent analysis and visualization in a dedicated dashboard.

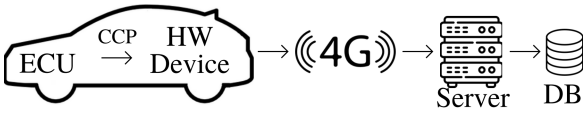


FIGURE 1. Communication diagram illustrating the main parts of the acquisition system.

The hardware communicates with the ECU through the Controller Area Network (CAN) bus, adhering to the CAN Calibration Protocol (CCP) for the Renault Sandero 2019. Sampling rates, limited to 4, 5, 10, and 100 ms, dictate the data acquisition process, with each experiment capable of capturing around 160 variables. The research strictly complies with bandwidth limits imposed by protocols during variable retrieval from the ECU.

A. FEATURE SELECTION

The same feature selection process detailed by [5] was applied in this extension, adding regression models, and recapitulated below:

Feature selection is crucial to address the limited number of variables per experiment and enhance the performance of machine learning (ML) models. This process reduces the set of variables representing the analyzed phenomenon. It prevents overfitting, eliminates redundant or noisy variables, and improves generalization, computational efficiency, and overall model performance [18], [43].

Various strategies were employed to create subsets of available attributes, assessing individual variables based on predefined criteria or direct sub-setting. A subsequent statistical evaluation determined the most qualified variables, with the selection process having a stopping condition and the best-performing subset prevailing until further evaluation [20]. Consequently, certain subsets of variables outperformed the full set and were adopted.

The feature selection specifically targeted engine subsystem variables related to fuel consumption. Utilizing algorithms like SelectPercentile, SelectKBest, SequentialFeatureSelector, SelectFromModel, RFE, and RFECV from the Scikit-Learn Python library [27], a total of 46 unique variables were chosen. These algorithms operate with supervised learning methods, ranking and selecting features through statistical measures with the target. Examples include SelectPercentile and SelectKBest, which rank features based on statistical measures, and SequentialFeatureSelector, SelectFromModel, RFE, and RFECV, which assess predictor performance using various feature combinations.

In this work, we applied feature selection for both classification and regression. In the classification context, the target variable was the consumption level label, and the predictors used for evaluation were the Logistic Regression and XGBoost Classifier models. For the regression problem, the target variable was the actual fuel consumption, and the evaluated models were Ridge, SVR, and XGBoost regressor. The selected variables and the methodology applied in this process will be presented in the following section.

B. DATA ACQUISITION

The collected data, influenced by selected variables and driving duration, undergoes real-time manipulation by workflows, providing runtime feedback to stakeholders or stored for subsequent offline analyses and reports through the database.

As explained by [5], experiments were conducted for 6 months, in varied environmental conditions, and aimed to explore subsets of the engine for comprehensive feature selection. Tests occurred exclusively within the Ágora Industrial Park in Joinville, Brazil, due to licensing constraints on public roads. The experiments were divided into 3 stages, each compiled in Table 3 and explained below, focusing on short-term driving scenarios that represent the day-to-day experiences of an average driver in non-traffic conditions, with a maximum duration of 40 minutes.

The first experiments were conducted from September to November and involved 30 acquisition variables taken from the related works, summarized as ''Features'' in Tables 1 and 2. In this phase, called Stage 1, we focus on understanding the relationship between variables and fuel consumption, achieved by running experiments divided into normal and aggressive driving conditions, making it evident in the data.

Subsequent experiments, represented as Stage 2, occurred from January to February. As many engine subsystems can influence fuel consumption, in this stage we selected over 220 variables from the most diverse types available in the ECU to help understand this relationship and to apply feature selection to improve training inputs. The variables tested ranged from boolean sensors that represent when a pedal is active or not, sensors that measure the angulation of

TABLE 3. Description of the acquisition datasets.

Stage	Total / Selected features	Duration of experiments (min)	Driver's behavior
Stg.1	30 / 13	13 - 27	Varied between aggressive and normal driving in the final half of the experiment
Stg.2	152 / 31	18 - 40	Varied between aggressive and normal driving throughout the experiment
Stg.3	141 / 28	10 - 15	Ordinary driver steering, not forced

TABLE 4. Fuel consumption - feature selection.

Engine Subsystem	Variable Description
Basic Component	Filtered engine speed
	Engine coolant temperature
	Engine speed
	Vehicle speed
	Total vehicle distance
	Accelerator pedal position
	Brake pedal - switches consolidation state
	Acceleration or deceleration vehicle state
	Relative throttle position in percent of the sensor supply voltage
	Value of applied offset correction
	Clutch pedal - minimum travel switch state - wire
	Instantaneous engine speed (tooth to tooth engine speed)
Air Control	Ambient air temperature
	Engine air load 1, 2, 3, and 4 TDC earlier
	Engine air load using predicted pressure
	Setpoint of the mass air flow pumped into cylinder
	Intake air temperature
	Throttle valve position setpoint
	Throttle valve position
	Intake manifold pressure
	Mass of air trapped in the cylinder for next ignition
Fuel And Richness Control	Final alcohol adaptive
	Injected fuel mass for ADAC [fuel consumption]
	Gas consumption for the mux
	Sum of fuel mass injected on the cylinder
	Injection time - cylinder 1, 2, 3, and 4 - 1st injection
	Catalyst warm-up is confirmed as necessary
	C factor
	Injected fuel mass setpoint disregarding cylinder
	Average injected fuel mass disregarding cylinder number or injection phase
	Effective time injection
	Offset applied on richness closed loop
	Richness close loop injection time factor applied
Ignition Advance	Maximum ignition advance
Engine Torque	Torque efficiency corresponding to minimum static ignition advance
	Estimated indicate engine torque
	Effective engine torque requested by real and virtual drivers
	Effective engine torque setpoint requested by real and virtual drivers
	Effective engine torque setpoint requested by real and virtual drivers
	Estimated effective engine torque
	Final torque target after modulation action
	Torque efficiency for idle speed controller
	Torque correction calculated
	Final torque setpoint after modulation action
Alternator and Battery Management	Battery voltage
	Alternator power

components, fluid injection, and distance traveled, to specific information such as adaptation factor on injection time, amount of air pumped into a cylinder, air conditioning request, engine air load using predicted pressure, torque efficiency corresponding to maximum ignition advance, fault vectors and sensors, among many others. These experiments forced the vehicle randomly during the route, not being so explicit in the data.

The final experiments, Stage 3, took place in February and were used to validate the ML models using variables selected by Feature Selection algorithms applied in the previous stages. In this phase, the vehicle was driven normally to simulate ordinary driving conditions.

The final variables found after the selection process are presented in Table 4, along with the subsystem of which it forms part.

IV. ALGORITHMS AND PARAMETERS DESCRIPTION

In this section, we describe the classification and regression algorithms and their related parameters. We chose the present algorithms after previous initial tests, where we reviewed the models present in the related works and literature, performed tests with the initially obtained data, and chose the ones that performed better with our available dataset, aside from covering a larger number of classes of algorithms, for comparison. Furthermore, we choose algorithms that are easy to explain to the general community, ensuring that AI systems provide human-understandable explanations for their decisions is crucial for gaining trust and acceptance in industrial applications [2].

A. CLUSTERS AND CLASSIFIERS

K-Means is an unsupervised clustering algorithm that divides the data into classes or clusters, with a number K of clusters informed as a parameter. In this method, the clusters are determined by centroids, K points in the data space, where a point belongs to a cluster if the respective centroid is the closest from it [1].

K-means determines its clusters through an iterative optimization process, starting with the K centroids initially randomized. In each step, the centroids are updated to the mean array in each cluster. Formally defining, given clusters $S_1, \ldots, S_K$, and points $x^{(1)}, \ldots, x^{(n)}$, it aims to minimize the total variance:

$$\sum_{i=1}^{K} |S_k| \sum_{x^{(i)} \in S_k} \|x^{(i)} - \overline{x}^{(k)}\|^2 \tag{1}$$

where $\overline{x}^{(k)} = (\overline{x}_1^{(k)}, \ldots, \overline{x}_m^{(k)})$ is the array with means of each input x_i respective to the k-th cluster. This is done by replacing the centroids in each step by the respective mean array, repeating it until the centroids no longer change positions, a process which has guaranteed convergence [12].

In this work, we used the Python SKLearn implementation of K-Means, choosing $K = 2$ as a parameter, as also used in related works [28], [40].

Logistic Regression Classifier is a supervised regression method used for classification, as the obtained function is a sigmoid, which returns real values within an interval between 0 and 1. Given inputs $x = (x_1, \ldots, x_m)$, the predictor have the form:

$$\hat{p}(x) = \frac{1}{1 + e^{-g(x)}} \tag{2}$$

where $g(x) = x^T\theta$, being $\theta = (\theta_1, \ldots, \theta_m)$ a parameter array. These parameters are obtained through the minimization of the loss function:

$$L(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y^{(i)}\log(\hat{p}^{(i)}) + (1 - y^{(i)})\log(1 - \hat{p}^{(i)})\right], \tag{3}$$

where $\hat{p}^{(1)}, \ldots, \hat{p}^{(n)}$ are the predicted values, $y^{(1)}, \ldots, y^{(n)}$ are the actual values (0 or 1). In the logistic regression, the predictor outputs probabilities, values in a range between 0 and 1, of the actual class being true. This can be used for binary classification, where the classes are determined if the function returns a value under or above 0.5 [1].

In this work, we used the Python SKLearn implementation of the Logistic Regression Classifier, with default parameters except for a higher max iterations number of 10000 steps in the training.

Gradient Boosting is a supervised method that joins classification and regression trees, predictors that classify data through certain binary conditions upon thresholds and value ranges to determine the class it belongs to, associating it with some predicted score. This method works by joining trees in sequence, where a tree is trained with the previous residual error, summing up their outputs to final results [1].

Each tree can be written as a predictor in the form $\hat{y}_k = f_k(x)$, where x is the inputs array, f_k represents the k-th tree, and $\hat{y}_k$ represents the output score. In ensemble methods, as in the case of Gradient Boosting, the final outputs are determined by:

$$\hat{y} = \sum_{k=1}^{K} f_k(x), \tag{4}$$

being $f_1, f_2, \ldots, f_K$ the trees obtained by the algorithm. Gradient boosting obtains the trees and their parameters θ through the minimization of an objective function in the form:

$$obj(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \omega(f_k). \tag{5}$$

In this objective function, the first sum is a loss term, where $x_1, \ldots, x_n$ are the train inputs, $y_1, \ldots, y_n$ the target values, $\hat{y}_1, \ldots, \hat{y}_n$ the predicted values, and $l(y, \hat{y})$ a convex loss function between predicted and actual values. The second sum is a regularization term, being $f_1, \ldots, f_K$ the tree structures and $\omega(f)$ some measure of tree complexity. What distinguishes Gradient Boosting from other ensemble methods is the use of a gradient-based approach, training each tree iteratively with the previous results. In this work, we used the XGBoost, optimized implementation of the Gradient Boosting, with its default parameters for training.

B. REGRESSORS

Ridge Regression is a type of regularized linear regression. It consists of a linear estimator trained with a regularized loss function, using the Ridge or Tikhonov Regularization technique, which aids in avoiding overfitting and influence from outliers [12]. Given inputs $\mathbf{x} = (1\ x_1\ x_2\ \cdots x_m)^T$, the technique aims to obtain a regressor in the form:

$$\hat{y} = \sum_{i=0}^{m} \theta_i x_i = \theta^T x, \tag{6}$$

where $\theta = (\theta_0\ \theta_1\ \cdots\ \theta_m)^T$ are numerical parameters. Given $x^{(1)}, \ldots, x^{(n)}$ train feature arrays and $y_1, \ldots, y_n$ train target values, these parameters θ are obtained through minimization of the Loss Function:

$$L(\theta) = \sum_{j=0}^{n} |y_j - \theta^T x^{(j)}|^2 + \frac{1}{2}\alpha\|\theta\|^2 \tag{7}$$

The first term in Equation 7 represents the usual least-squares linear regression loss function and the term $\frac{1}{2}\alpha\|\theta\|^2$ is called regularization term, being $\alpha > 0$ a fixed parameter and $\|\theta\| = \sqrt{\sum \theta_i^2}$ the usual Euclidean Norm.

In this work, we use the Python Sci-kit Learn implementation of Ridge Regression, with the default parameters set in the library.

Support Vector Regression (SVR) is a method that implements ideas from the Support Vector Machines into regression problems. The final estimator is in the form $\hat{y} = \theta_0 + \sum_{i=1}^{m}\theta_i x_i = \theta^T x$, being $x = (1\ x_1\ \cdots x_m)$ the input, $\hat{y}$ the predicted value and $\theta = (\theta_0\ \cdots\ \theta_m)$ an array of numerical parameters. In this model, instead of fitting an estimator which minimizes the error $|y - \hat{y}|$, the algorithm tries to fit an estimator such that the maximum possible inputs are in a specified $\varepsilon > 0$ error margin: the estimator fits a maximum quantity of $x^{(j)}$ inputs respecting $|y_j - \theta^T x^{(j)}| \leq \varepsilon$. The θ parameters are obtained by solving the minimization problem:

$$\min_{\theta,\zeta} \quad \frac{1}{2}\|\theta\|^2 + C\sum \zeta_i \tag{8}$$

$$\text{s.t. } |y_j - \theta^T x^{(j)}| \leq \varepsilon + \zeta_j, \tag{9}$$

where $C > 0$ is a fixed regularization parameter and $\zeta_i \geq 0$ are called slack variables, implemented to consider margin violation points.

In this work, we use the Scikit-Learn implementation "LinearSVR", which implements the SVR optimized for the linear estimator case. We used the default parameters set by the library, except for $\varepsilon = 1$.

Gradient boosting, the same used for the classification task, was adapted for the regression. It applies the same principle of tree ensembles and training, but instead of a score

related to a classification task, the prediction consists of the actual value for regression itself [6].

For this work, we use the XGBoost implementation of Gradient Boosting, which uses computational and mathematical resources for optimizing both training and prediction tasks.

Nonlinear AutoRegressive with Exogenous Inputs Recurrent Neural Network (NARX-NN) is a type of autoregressive model, which uses previous predictions in addition to other features and its previous values to evaluate the future ones, integrated into Neural Networks architectures. NARX consists of an autoregressive model that supports nonlinear estimators and the use of other features beyond the actual time series values, in this context called Exogenous Inputs, as described by [16].

Given a time series with values $y_1, \ldots, y_T$, exogenous values $x^{(1)}, \ldots, x^{(T)}$ with each $x^{(i)}$ having inputs on the form $x^{(i)} = (x_1^{(i)}, \ldots, x_m^{(i)})$, and a nonlinear estimator f, the NARX model is a predictor in the form:

$$\hat{y}_{T+1} = f(\hat{y}_T, \hat{y}_{T-1}, \ldots, \hat{y}_{T-N}, x^{(T+1)}, \ldots, x^{(T-N)}), \quad (10)$$

where N is called the lag or order of autoregression, which specifies the number of previous inputs in use, and $\hat{y}_i$ are the series predicted values. The NARX Recurrent Neural Network (NARX-NN), as described by [16] consists in the use of a regressor neural network as the f estimator. The training of this model consists of the usual training of the f, considering all the listed inputs and setting the actual series values as the training autoregressive inputs.

We use the Python FireTS library implementation of NARX, which provides support for any type of nonlinear estimator integration. For the selected neural network, we used the Sci-kit Learn MLP Regressor implementation, which consists of a Multilayer Perceptron Neural Network optimized for regression tasks.

To select the best parameters for this model, we evaluated it using a different combination of lag and number of neurons for the hidden layer as done by [16], with the number of neurons varying between 30 and 110 in steps of 5, and lag varying between 2 and 7. The best results, among different tests with the data, were with the NARX-NN having 75 neurons and a lag of 5.

For the implementations, we used the Python version 3.8.10, the Sci-kit Learn library version 1.2.1 (for the models where it is mentioned), and the XGBoost version 1.7.0. All algorithms are available online at https://github.com/canalrafael/Fuel-Consumption-Analysis.git.

V. EVALUATION

In the following subsections, we present the results obtained in the classification of the driver's driving profile and in the prediction of fuel consumption. Additionally, we make comparisons with related works that are close to ours (described in Section II). Subsection V-A presents the results already discussed in the article "Driving Profile Analysis Using Machine Learning Techniques and ECU Data" [5] replicated here for a better understanding of the research extension.

A. CLASSIFIER RESULTS

To evaluate the classifiers, we performed cross-validation, partitioning the datasets into training and testing sets. The chosen partitioning method consisted of taking one experiment of a "Stage" as a test set and the others for training, similar to [29]. This technique allows us to verify the generalization of the models, in addition to applying in results the metrics of accuracy, precision, and recall, as used and described by [28]. In this way, we obtained the overall performance of the models and how many correct classifications were made, visualizing the number of false positives and negatives.

Figure 2 shows a graph of a test experiment with the fuel consumption variable (blue) highlighted by the High Consumption class (red) made by XGBoost. The points not marked with color represent the economic consumption class. In this example, the model identifies whether the consumption level was high or low, compared to the vehicle's original consumption variable.

As explained in Section III-B, we divided the experiments into Stages, in which we applied changes in the selected variables due to the knowledge acquired (i.e., domain-specific) and the feature selection process. Table 5 summarizes the results, progressively (stage after stage), making explicit the benefits of a better understanding of the problem and the implications of feature selection algorithms.

Table 5 also shows the training and testing results obtained by the explored classification and clustering methods, through the different experimental stages. The shown metrics do not bring all detailed evaluations, but a summary of gathered results from different feature sets, as well as experiments train and test divisions, showing the range, average, and standard deviation of the obtained scores for different scenarios.

When comparing the experiment stages, Stage 1 shows lower performances, ranging from 80% to 95% of accuracy, while the second and third stages show better results, varying from 81% up to 99% of accuracy. Furthermore, the precision of the latest experiments is far higher, reaching up to 100% scoring, yet a smaller recall is obtained. This shows an improvement in the updated variable set for classifying levels of consumption, but still with good results with the older set. Analyzing the distributions and value range of consumption records for some experiments, there were cases where a lower variety in the data, which combined with the wrong selection of features, led to a weaker performance. Overall, the obtained results showed that the classification methods had good precision in determining the consumption level from the right selected features, reaching up to 100% of scoring for all metrics in some of the train and test splits.

Regarding the classifier methods, in general, XGBoost shows a slightly better scoring than the Logistic Regression. This changed in some stages and for certain features selected

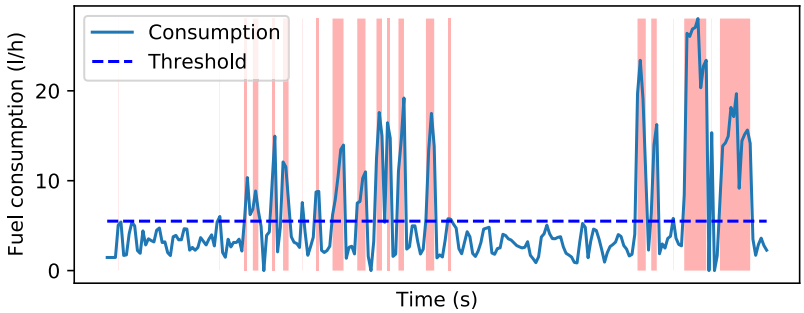


FIGURE 2. Fuel consumption (in blue) over time, with the classification of high consumption periods (in red).

TABLE 5. Classification results evaluation metrics.

	Algo.	Info.	Accuracy	Precision	Recall
Stage 1	XGBoost	Range	80.60%-94.80%	76.30%-97.40%	72.30%-99.40%
		Average	88.09%	85.43%	89.98%
		Std dev.	5.47%	6.26%	10.22%
	Logistic Regre.	Range	20.39%-92.72%	29.38%-100.00%	0.61%-100.00%
		Average	75.63%	78.86%	77.34%
		Std dev.	22.63%	20.23%	28.23%
	K-Means	Range	26.29%-79.31%	29.27%-100.00%	46.60%-53.40%
		Average	66.63%	75.30%	52.13%
		Std dev.	18.16%	21.77%	2.48%
Stage 2	XGBoost	Range	81.21%-99.59%	92.82%-100.00%	40.54%-98.53%
		Average	95.42%	96.86%	88.07%
		Std dev.	6.41%	2.77%	21.27%
	Logistic Regre.	Range	70.64%-97.93%	53.52%-100.00%	35.14%-95.68%
		Average	88.94%	86.17%	78.44%
		Std dev.	9.33%	15.71%	22.71%
	K-Means	Range	6.45%-93.55%	7.94%-98.07%	19.76%-91.70%
		Average	60.41%	52.32%	61.92%
		Std dev.	31.36%	30.63%	24.53%
Stage 3	XGBoost	Range	95.00%-100.00%	93.33%-100.00%	93.33%-100.00%
		Average	97.30%	96.82%	96.11%
		Std dev.	1,00%	3.18%	3.29%
	Logistic Regre.	Range	85.00%-100.00%	76.47%-100.00%	80.00%-100.00%
		Average	92.67%	90.92%	90.56%
		Std dev.	10.13%	14.11%	11.85%
	K-Means	Range	6.83%-90.68%	6.85%-100.00%	8.33%-75.00%
		Average	45.65%	45.42%	39.72%
		Std dev.	34.05%	39.93%	23.83%

sets, but overall, XGBoost had more persistent performances, ranging between high averages and low standard deviations. Besides, it had better prevalent results, as highlighted in green in Table 5, showing greater average scores, and lower variations, showing persistence in them. In contrast, Logistic Regression shows higher scores in some cases, but also has lower averages and higher standard deviations on the results, reaching accuracies and recalls under 30% for some specific tests and feature sets, which did not occur with XGBoost, although had fair results in general.

The K-Means clustering presented inferior results in comparison with the classification algorithms, trained specifically for the given classes. It got lower accuracy, precision, and recall averages in all steps, ranging from 40% to 75%. The reason can be found in visualizing the standard deviation in Table 5, with values greater than the ones obtained in the other algorithms. K-Means proved to be more sensitive to the set of variables used in training, having precision values below 10%, bringing the average down. Even with this issue of having wider variations with the wrong resource sets that negatively affect its performance, in general, K-means was capable of getting a reasonable score, showing accuracy, precision, and recall up to 90% in some tests. This indicates that the specific threshold used for comparison was not perfect or that it can be adjusted for the identified clusters, reinforcing that the driving styles clusters given by K-Means may also capture a division in consumption levels, not only a binary division. It is also notable that no specific feature selection was applied for K-Means, using for some tests the same features selected for the classification algorithms and the related works common features, which also have contributed to lower and less persistent results. Besides, in our work, we performed the comparison of the clusters with real data of the instantaneous consumption levels, in contrast to [28] and [44], where simulated data or no fuel consumption were observed.

Compared to related works, we improved the overall results, having metrics with superior values, an extensive feature selection work that accurately identifies the driver's relationship with key variables of fuel consumption, and ensuring the integrity of the data used. One of the observed advances, when compared to similar models and analysis [29], is obtaining similar or superior metrics using lower-cost methods. In the referred work, the authors got an accuracy of 80% to 83% with more complex methods such as Neural Networks, using visual information in addition to the vehicle-gathered data, while in this work, by using only vehicle-observed features, we reach 100% accuracy in some scenarios through computationally less expensive models as Logistic Regression [8] and Gradient Boosting [34], besides they are not a black-box. This would allow the execution of classification within the ECU and not on a cloud server, for instance.

Furthermore, in this work, we predicted a consumption level with high accuracy and precision using directly a level of measured consumption as a basis. Similar works obtained equivalent or lower accuracy and precision, from 80% to 100%, using labels generated by clusters for classification, as in [28] and [44], while in this work we managed to obtain the same or better performance, up to 90% and 100%, using direct reference consumption for training and testing classifiers.

B. REGRESSION RESULTS

To evaluate the predictive models for fuel consumption various metrics can be used, such as mean squared error (MSE), mean absolute error, root mean squared error, and coefficient of determination (R^2), among others [45]. To allow a better comparison with related works and, because they complement each other in the analyses, we evaluated the following ones in each set of algorithms: MSE, R^2 score, and MAE.

Mean Squared Error (MSE) is a metric that shows an average of the square of errors between predicted and measured values, using the squared term that gives higher penalties for higher error values. For each set of measured values $y_1, \ldots, y_n$ and predicted values $\hat{y_1}, \ldots, \hat{y_n}$, the MSE is determined by

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y_i})^2. \tag{11}$$

Mean Absolute Error (MAE) is a metric that measures the average of the errors using the absolute error value, which stands for a slightly moderate average, without distinction from higher or lower penalties. From some set $y_1, \ldots, y_n$ of measured values and $\hat{y_1}, \ldots, \hat{y_n}$ predicted values, the MAE is obtained by:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y_i}|. \tag{12}$$

R^2 is a metric that evaluates how much of the data variance is explained by the predicted values given by the model. It has a maximum value of 1, where the score closer to 1 represents a better fit. Given $y_1, \ldots, y_n$ real values, $\bar{y}$ their mean value and $\hat{y_1}, \ldots, \hat{y_n}$ predicted values, R^2 is obtained by the following equation

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y_i})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{13}$$

To assess the regression algorithms, we executed a similar analysis with the same cross-validation procedure as explained in the previous subsection. For the first two Stages, we separated in each round one of its experiments for testing and the remaining for training. For the third Stage, we performed 3-fold cross-validation, where data was randomly split into 3 sets, where we performed cross-validation between them selecting one for test and the remaining for training.

To illustrate, Figure 3 shows the predicted fuel consumption levels estimated by the XGBoost Regressor (in orange) compared to the actual measured consumption (in blue). The graph exemplifies the results given by these regressor models, which provide predictions of the actual values of fuel consumption levels in each instant.

Similar to the classification evaluations, we summarize the results of regression models in Table 6, aggregating different train and test results by Stage of experiments and used algorithm.

Results in Table 6 show a wider range of obtained scores and metrics between models and Stages of experiments, using different sets of experiments. Concerning the Stages, once more we observe overall lower scores for the Stage 1 experiments, an improvement for Stage 2, and the best scores for Stage 3 experiments. In Stage 1 results, it can be seen for example an average R^2 of 0.78 in XGBoost Regression and 0.62 with SVR, besides a higher MSE average of 4.49 and 6.31. The MSE is slightly higher than MAE, with 0.73 for XGBoost and 1.27 for SVR, indicating that these averages are due to high peak errors, with an overall good score in most cases. Still, we obtained some high outlier errors, as with Ridge Regression, which occurred using some specific feature and experiment sets.

In Stage 2, a similar case occurs, with similar R^2 and error averages for XGBoost and SVR, besides the higher R^2 peaks of 0.97 for XGBoost, 0.95 for SVR, and 0.97 in Ridge, as can be seen in the range values of these metrics. Yet, in Stage 3, we could obtain some overall good results, with R^2 averages above 0.9 and MSE average scores below 1 (l/h), as highlighted in Table 6. This shows once more the improvement of the newly available features in the latest experiment Stages, shown by the higher and more consistent scores obtained in Stage 3.

Concerning the models, the XGBoost was the one with the most consistent and good results in general for all stages and feature sets. As may be seen in Table 6, it kept similar results in Stages 1 and 2, being the better one in the

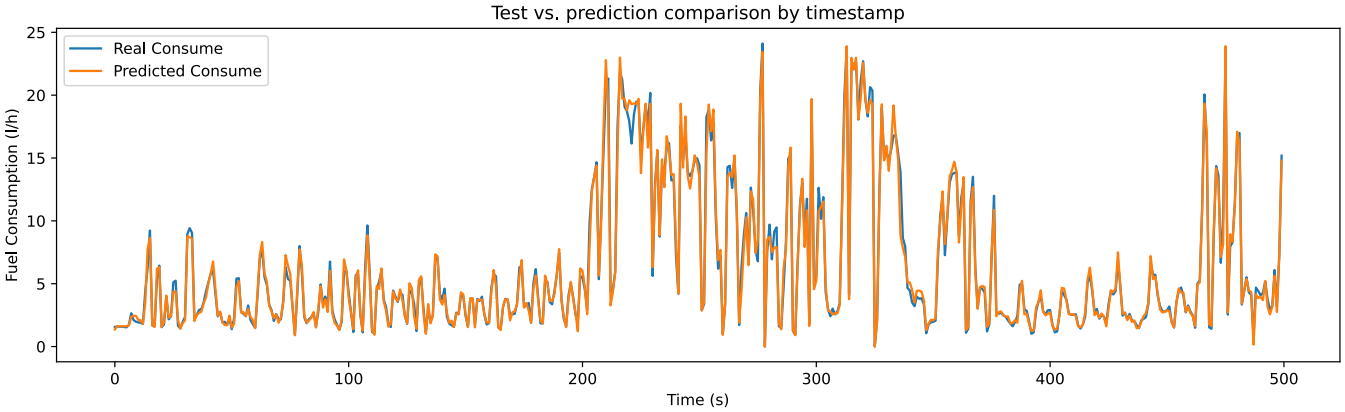


FIGURE 3. Comparison between measured (blue) and predicted (orange) fuel consumption levels over an experiment, through XGBoost regressor.

TABLE 6. Regression results evaluation metrics.

	Algo.	Info.	MSE ((l/h)^2)	R^2	MAE (l/h)
Stage 1	XGBoost	Range	0.89-11.73	0.54-0.91	0.55-1.02
		Average	4.49	0.78	0.73
		Std dev.	3.64	0.12	0.17
	Ridge Regres.	Range	2.58-4255.82	-434.42 -0.84	0.97-65.19
		Average	414.35	-40.58	9.55
		Std dev.	1219.00	124.99	18.66
	SVR	Range	3.36-16.66	0.35-0.81	0.96-2.09
		Average	6.31	0.62	1.27
		Std dev.	3.53	0.14	0.32
Stage 2	XGBoost	Range	0.93-16.12	0.49-0.97	0.46-1.74
		Average	7.41	0.77	1.01
		Std dev.	6.79	0.22	0.52
	Ridge Regres.	Range	0.94-164.08	-4.19-0.97	0.61-9.88
		Average	26.40	0.17	2.66
		Std dev.	45.52	1.44	2.53
	SVR	Range	1.63-15.41	0.51-0.95	0.79-2.25
		Average	7.01	0.78	1.39
		Std dev.	4.94	0.16	0.46
Stage 3	XGBoost	Range	0.37-0.68	0.96-0.99	0.25-0.42
		Average	0.49	0.98	0.32
		Std dev.	0.08	0.01	0.05
	Ridge Regres.	Range	0.28-1.80	0.75-0.99	0.23-1.35
		Average	0.91	0.91	0.66
		Std dev.	0.57	0.09	0.42
	SVR	Range	0.38-0.71	0.95-0.99	0.30-0.58
		Average	0.58	0.97	0.44
		Std dev.	0.10	0.01	0.08

first Stage, showing MAE and R^2 averages above 0.9 and below 0.9 respectively, with just slightly lower mean scores than SVR for Stage 2. Furthermore, Stage 3, has shown overall better performance in comparison to all models (R^2 scores above 0.96, average MSE and MAE respectively below 0.5 (l/h)^2 and 0.4 (l/h), among with consistent results shown by the small deviation in the scores). This indicates the model's capability to get overall good predictions in most cases. In comparison, SVR was also capable of showing good results, reaching up to 0.95 and 0.99 R^2 scores in Stages 2 and 3, and the average MSE and MAE not above 7 (l/h)^2 and 1.4 l/h respectively. Finally, the Ridge Regression model has shown by far the lowest performances, in some cases having far higher errors, as in Stages 1 and 2, with negative R^2 and extreme MSE and MAE values. This shows that, in general, yet having some good scores in some cases as the highlighted MAE and MSE range scores in Stage 3, it may not generalize well for most cases.

Regarding the results of the NARX-NN, a model that was adapted from [16], we performed tests with different sets of features and settings for the model, using the same proceeding and similar settings described by the authors with some adjustments for our data. However, the algorithm did not show reasonable scores in comparison to the other explored algorithms, not being possible to replicate the results achieved in their application. Among all the validations, the higher R^2 score obtained was below 0.21, with the majority of the cases getting negative scores, besides the high MSE and MAE, showing that the model did not perform well with our data.

In comparison to the related works, we were able to achieve results similar or better to other explored models. For example, in [11] and [24] similar algorithms were explored for the regression problem getting an R^2 score of 0.96 and 0.97, while we could obtain up to 0.99 of R^2 in the observed tests with XGBoost and SVR. It is noticeable that in our case the consumption data was provided by ECU sensors, instead of the use of physical models or database-provided data. Furthermore, the presented results show advancements also in comparison to [42], which using even more complex Neural Network models reached R^2 scores from 0.5 to 0.6, while using computationally less expensive models we could reach scores between 0.8 and 0.99, besides of an MSE of 0.49 (l/h)R^2, corresponding to an RMSE of 0.7 l/h in comparison to the 0.87 l/100km RMSE obtained in their work. Finally, although we applied similar tests and settings, as mentioned above, the model explored by [16] was not able to show good results in our dataset. Still, with less expensive models as XGBoost and SVR, we were able to reach an average MSE of 0.49 (l/h)R^2 and 0.58 (l/h)R^2 respectively, or even down to 0.28 in Ridge Regression and 0.37 in XGBoost, compared to the 0.43 (kpl)R^2 obtained by these authors with use of the Recurrent Neural Network.

VI. CONCLUSION

Implementing Industry 4.0 technologies such as IoT and AI can enhance automation and efficiency in automobile manufacturing processes. This can lead to improved production quality, reduced lead times, and increased flexibility in responding to market demands.

The presented work analyzed two scenarios related to fuel consumption. First, we compared three ML models to classify the driving profile concerning fuel consumption, and, second, compared three ML regression models to predict the fuel consumption, proving its validity on both subjects with results that indicate a correct categorization of the driver in economic or non-economic and an accurate prediction of consumption. The evaluation of the models (training and testing phases) was carried out using real data collected from a vehicle ECU. The models were integrated into a cloud server, which allows for data processing at runtime (while the car is running), making it possible to monitor the results using web tools or local software applications. In this way, drivers and stakeholders can monitor vehicle data while driving, making the work applicable in various sectors such as industry, security, tracking, transport, and automotive in general.

In the classifiers, we could obtain values up to 100% of accuracy, precision, and recall, with an average of around 97% in models like XGBoost, as well as similar results with Logistic Regression. This shows an improvement to other works, where lesser scoring was obtained, despite using more complex and expensive models or not real collected data. Furthermore, in the regressors our models were able to show R^2 scores up to 0.99, an MAE of 0.23 (l/h), besides an average MSE of 0.49 (l/h)^2, reaching down to 0.28 (l/h)R^2 in some tests. These results, compared to similar works, also show advancements in reaching good scores with similar or even less expensive models, besides the use of real data collected for the training and tests.

Due to the multidisciplinarity of the automotive industry and the demand for more data, we expect vehicle ECUs and other automotive components to generate more information, creating the opportunity for more complex data-based analyses with ML.

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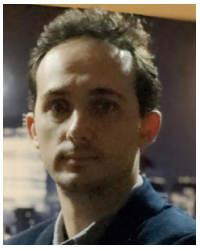
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